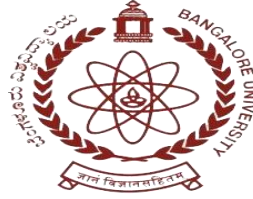


BANGALORE UNIVERSITY
JNANABHARTHI CAMPUS, BANGALORE - 560056



INTERNSHIP REPORT ON
“SPATIO-TEMPORAL LAND USE LAND COVER ANALYSIS OF MYSURU DISTRICT USING MACHINE
LEARNING AND STATISTICAL TECHNIQUES (2018-2024)”

Submitted in partial fulfillment of the requirement for the award of the degree

Of

Bachelor of Science

in

Geography and Geoinformatics

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Internship carried out at:

Karnataka State Remote Sensing Applications Centre (KSRSAC) Bangalore

Guide:

Mrs. Vidya J & Mr. Srikanth



KARNATAKA STATE REMOTE SENSING APPLICATION CENTER

(KSRSAC) – VTU Extension Centre

DPAR (e-Governance), GOK

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2025 - 2026

DECLARATION

We **Ria Dutta U03NK23SS0040**, **Naval Kishore U03NK23S0044**, **Riyon Das U03NK23SS0050**, **Rubeena R Sone U03NK23S0064** hereby declare that this dissertation work titled “**SPATIO-TEMPORAL LAND USE LAND COVER ANALYSIS OF MYSURU DISTRICT USING MACHINE LEARNING AND STATISTICAL TECHNIQUES (2018-2024)**” was independently carried out by us.

This Internship report is submitted to **Bangalore University**, Bengaluru, Karnataka in partial fulfilment of the requirements for the award of the Degree of Bachelor of Science in Geography & Geoinformatics, during the year 2025-2026.

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Place: Bengaluru

Submission Date: 25.05.2026

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CHAPTER 1



Fig 1: KRSAC Bangalore

1.1 ABOUT THE INSTITUTION

The Karnataka State Remote Sensing Application Centre (KRSAC) is a well-established geospatial organization functioning under the Department of Personnel and Administrative Reforms (DPAR), Government of Karnataka. The Centre focuses on the application of remote sensing, Geographic Information Systems (GIS), Global Positioning Systems (GPS), and related geospatial technologies to support planning, monitoring, and decision-making activities in different sectors of governance. Established with the objective of integrating modern spatial technologies into administrative systems, KRSAC has become an important technical institution providing geospatial solutions to government departments, research institutions, and public agencies. Over the years, the relevance of geospatial technology has increased rapidly due to the growing need for accurate, real-time, and dependable data for developmental activities. KRSAC plays an important role in connecting satellite-based data collection with field-level implementation by transforming raw spatial information into meaningful insights. The center provides extensive support in areas such as land resource management, environmental monitoring, infrastructure planning, disaster management, digital governance and contributes towards improving efficiency, transparency, and accountability in public administration.

CHAPTER 2

TASK PERFORMED

2.1 INTRODUCTION

Land Use Land Cover (LULC) analysis is essential for understanding environmental changes caused by urbanization, agriculture, and human activities. Remote sensing and GIS techniques provide effective methods for monitoring these changes over time. With the advancement of cloud-based platforms like [Google Earth Engine](#), large volumes of satellite data can be processed efficiently for accurate LULC mapping and change detection.

Mysore District has experienced significant land use changes due to population growth, urban expansion, and agricultural development. This study focuses on analysing the spatiotemporal LULC changes in Mysore District from 2018 to 2024 using satellite imagery. Machine learning algorithms such as Random Forest (RF) and Support Vector Machine (SVM) are used for image classification, while [ERDAS IMAGINE](#) is utilized for image processing and change detection analysis.

2.2 OBJECTIVES

The primary objective of this study was to analyse the spatio-temporal Land Use Land Cover (LULC) changes in Mysore District from 2018 to 2024 using multi-temporal satellite imagery. The study focused on understanding the transformation of major land cover classes over time and assessing the spatial patterns of landscape change within the district.

Another key objective was to perform supervised classification of satellite imagery using Random Forest (RF) and Support Vector Machine (SVM) algorithms in Google Earth Engine (GEE). The study also aimed to compare the classification performance of these machine learning techniques with the statistical classification methods implemented in ERDAS IMAGINE 2014 using standard accuracy assessment parameters.

Additionally, the study aimed to identify significant land cover transformations such as urban expansion, vegetation loss, and changes in agricultural land during the study period. This analysis helps in understanding the impact of urbanisation and land use changes on the regional environment and landscape dynamics.

2.3 METHODOLOGY

The methodology adopted for the spatio-temporal Land Use Land Cover (LULC) analysis of Mysore District involved the use of both Google Earth Engine (GEE) and ERDAS IMAGINE 2014 to

compare machine learning-based and statistical supervised classification techniques for the years 2018, 2021, and 2024. Initially, the Area of Interest (AOI) boundary of Mysore District was obtained and used to clip the satellite imagery for each study year. Satellite datasets were prepared using False Colour Composite (FCC) and Natural Colour Composite (NCC) images to enhance visual interpretation and class separability. In Google Earth Engine, Sentinel satellite imagery corresponding to the selected years was imported and pre-processed using cloud filtering and compositing techniques. For ERDAS IMAGINE, the required spectral bands were stacked and exported from GEE for further processing. A classification scheme consisting of seven LULC classes, namely Built-up, Agriculture, Forest, Grassland, Wasteland, Wetlands, and Waterbodies, was developed to maintain consistency across all years and platforms.

Training samples representing each LULC class were collected through visual interpretation of the imagery and used for supervised classification. In GEE, machine learning classifiers such as Random Forest (RF) and Support Vector Machine (SVM) were trained using the collected samples to generate classified LULC maps for each study year. In ERDAS IMAGINE, supervised classification was performed using statistical classification techniques through the creation of signature files and training datasets. The classified outputs obtained from both platforms were exported for further visualization and comparative analysis using GIS and remote sensing tools. To evaluate the reliability and performance of the classifications, accuracy assessment was carried out using confusion matrices and statistical measures including Producer's Accuracy, User's Accuracy, Overall Accuracy, and Kappa Coefficient. These assessments enabled a comparative evaluation of the classification efficiency and accuracy between the machine learning approaches implemented in GEE and the statistical classification techniques applied in ERDAS IMAGINE.

2.3.1 GOOGLE EARTH ENGINE (GEE)

1. Study Period Selection - Selected the years 2018, 2021, and 2024 for spatio-temporal LULC analysis of Mysore District.

2. Data Preparation- Imported satellite image layers including False Color Composite (FCC), Natural Color Composite (NCC), and AOI boundary into [Google Earth Engine](#).

3. Training Sample Collection

Collected training samples for seven LULC classes:

- Built-up
- Agriculture
- Forest
- Grassland
- Wasteland

- Wetlands
- Waterbodies

4. Classification Process

- Trained the samples using the Random Forest classification and SVM model.
- Executed the classification to generate LULC maps for each study year.

5. Data Export and Analysis

- Exported the classified images to Google Drive.
- Performed further analysis and visualization using [ERDAS IMAGINE](#) and GIS tools.

6. Accuracy Assessment - Performed classification accuracy evaluation using:

- Confusion Matrix
- Producer's Accuracy
- User's Accuracy
- Overall Accuracy
- Kappa Coefficient

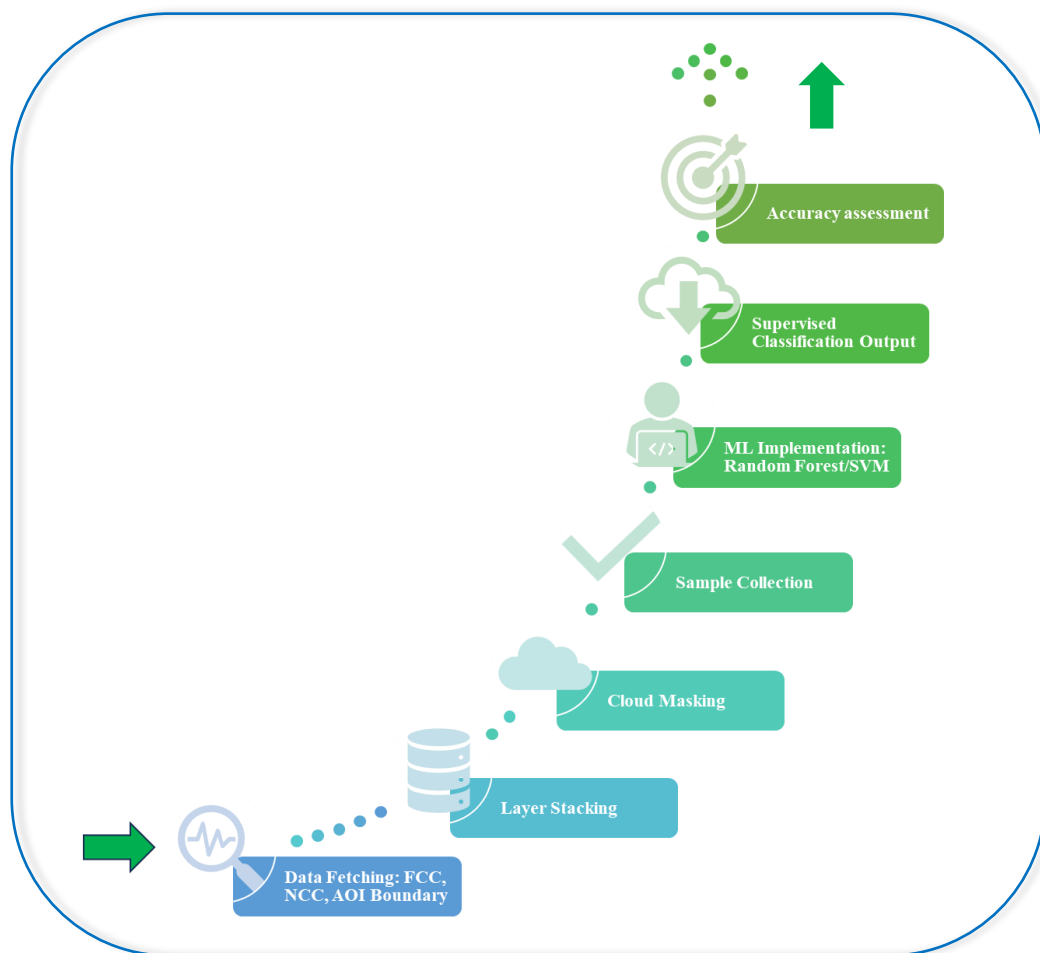


Fig 2: GEE Methodology Workflow

2.3.2 ERDAS IMAGINE

1. Study Period Selection – Selected the years 2018, 2021, and 2024 for spatiotemporal LULC analysis of Mysore District.
2. Image Processing and Band Stacking – Selected and stacked spectral bands (B8, B4, and B3) in [Google Earth Engine](#) for analysis.
3. Data Export – Exported the stacked layer to Google Drive for further processing in GIS/remote sensing software.
4. Classification Scheme –

Defined seven Land Use Land Cover (LULC) classes:

- Built-up
- Agriculture
- Forest
- Grassland
- Wasteland
- Wetlands
- Waterbody

5. Signature File and Sample Collection – Created a signature file and collected training samples for each LULC class for supervised classification.
6. Accuracy Assessment -

Performed classification accuracy evaluation using:

- Confusion Matrix
- Producer's Accuracy
- User's Accuracy
- Overall Accuracy
- Kappa Coefficient

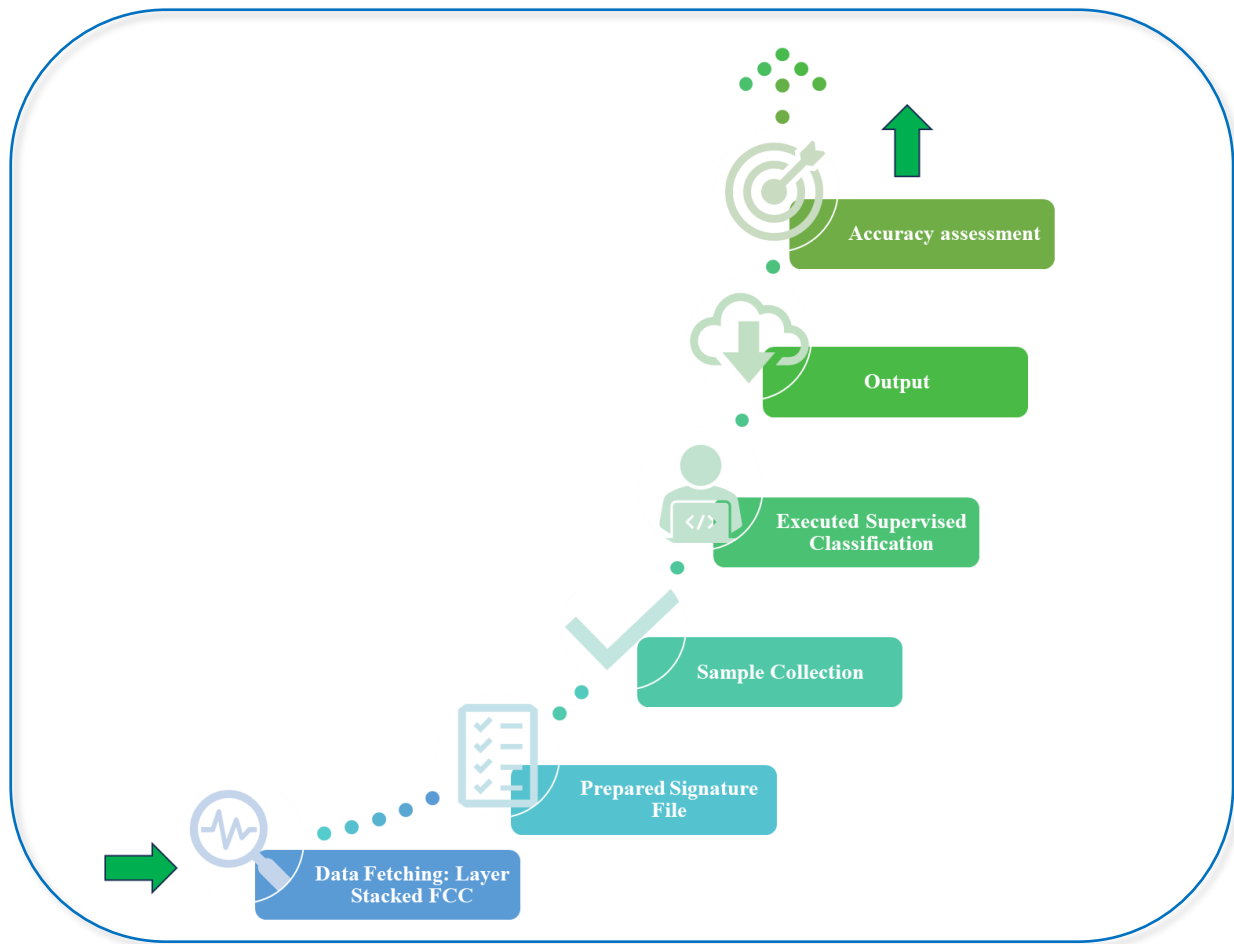


Fig 3: ERDAS Imagine Methodology Workflow

2.3.3 VALIDATION TECHNIQUES

Accuracy assessment is an essential step in supervised classification as it evaluates the reliability and correctness of the generated Land Use Land Cover (LULC) maps. In this study, the classified outputs obtained from Google Earth Engine and ERDAS IMAGINE were validated using statistical accuracy assessment techniques such as Overall Accuracy, Kappa Coefficient, Producer's Accuracy, User's Accuracy, and the Confusion Matrix. These techniques compare the classified image with reference or ground truth data to determine how accurately each land cover class has been identified. The Confusion Matrix, also known as the Error Matrix, forms the basis for most accuracy assessment parameters. It is a square matrix that compares the actual reference classes with the classified classes and indicates correctly and incorrectly classified samples. The diagonal elements of the matrix represent correctly classified samples, while off-diagonal elements represent classification errors. From this matrix, Producer's Accuracy and User's Accuracy can be derived for each class.

Producer's Accuracy measures the probability that a reference pixel has been correctly classified and is calculated by dividing the number of correctly classified pixels in a class by the total number of reference pixels for that class. It mainly indicates omission error, which occurs when pixels belonging

to a particular class are excluded from that class during classification. User's Accuracy, on the other hand, measures the probability that a pixel classified into a particular category actually represents that category on the ground. It is calculated by dividing the number of correctly classified pixels in a class by the total number of pixels classified into that class, thereby indicating commission error. Overall Accuracy represents the total proportion of correctly classified samples and is calculated by dividing the sum of correctly classified pixels by the total number of reference samples. It provides a general measure of classification performance but does not account for agreement occurring by chance. Therefore, the Kappa Coefficient is additionally used to measure the agreement between classified and reference data while considering random agreement. Kappa values range from 0 to 1, where values closer to 1 indicate strong agreement and higher classification reliability. Together, these validation techniques provide a comprehensive evaluation of the performance and accuracy of the supervised classification methods used in the study.

2.4 RESULTS AND DISCUSSIONS

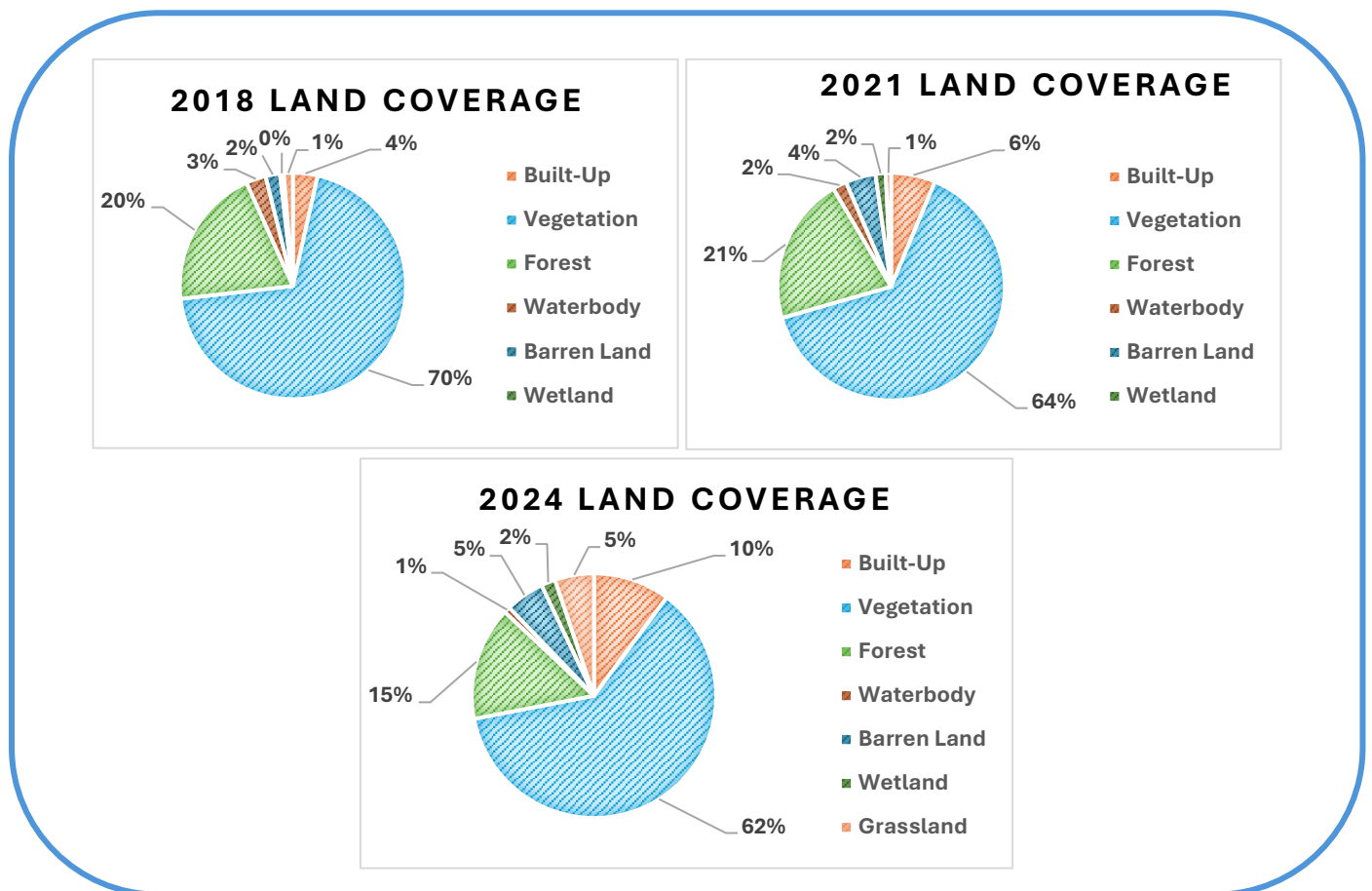


Fig 4: Area coverage (sq. km)

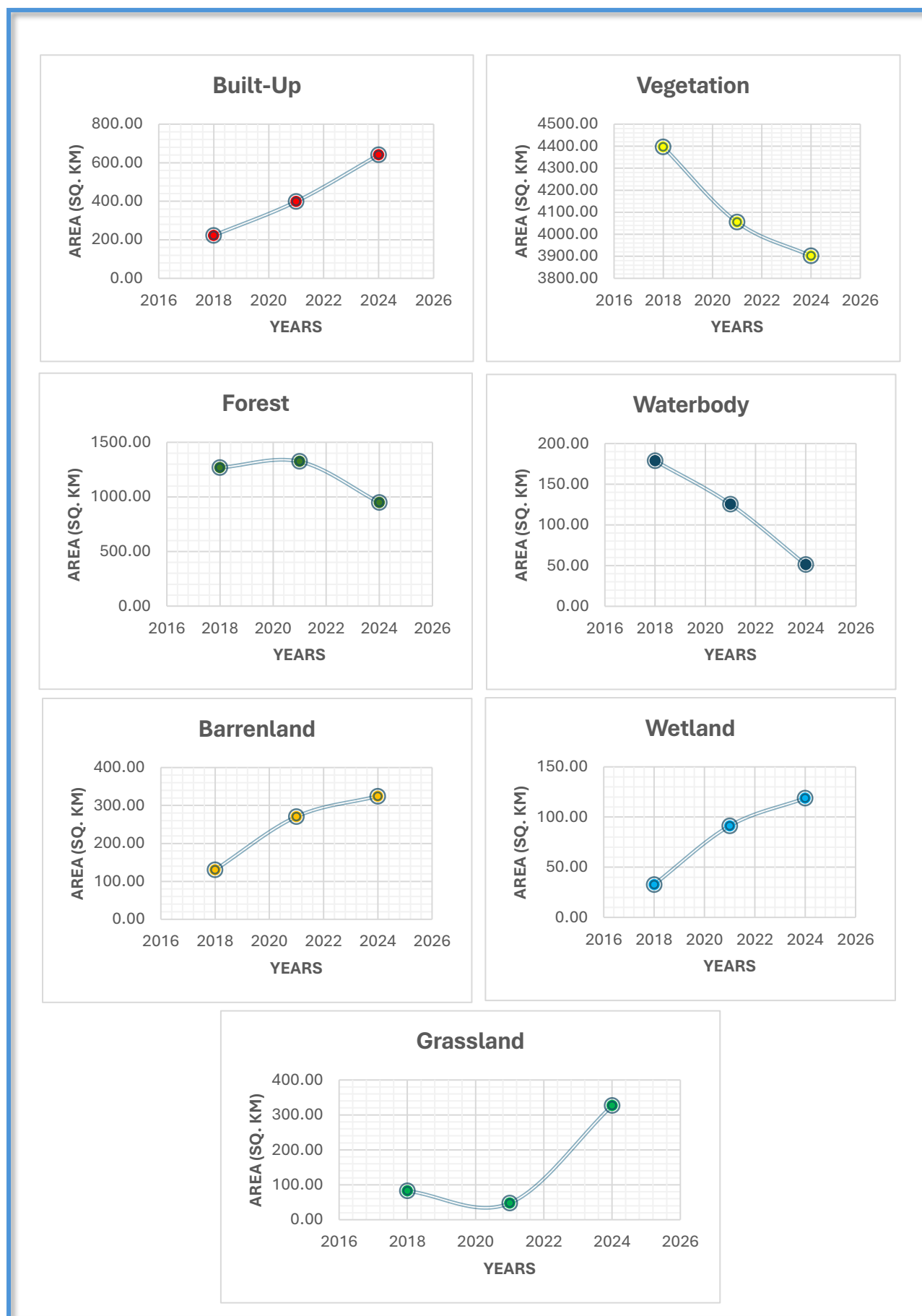


Fig 5: Temporal analysis of classified areas

SUPERVISED CLASSIFICATION USING RANDOM FOREST (GEE)

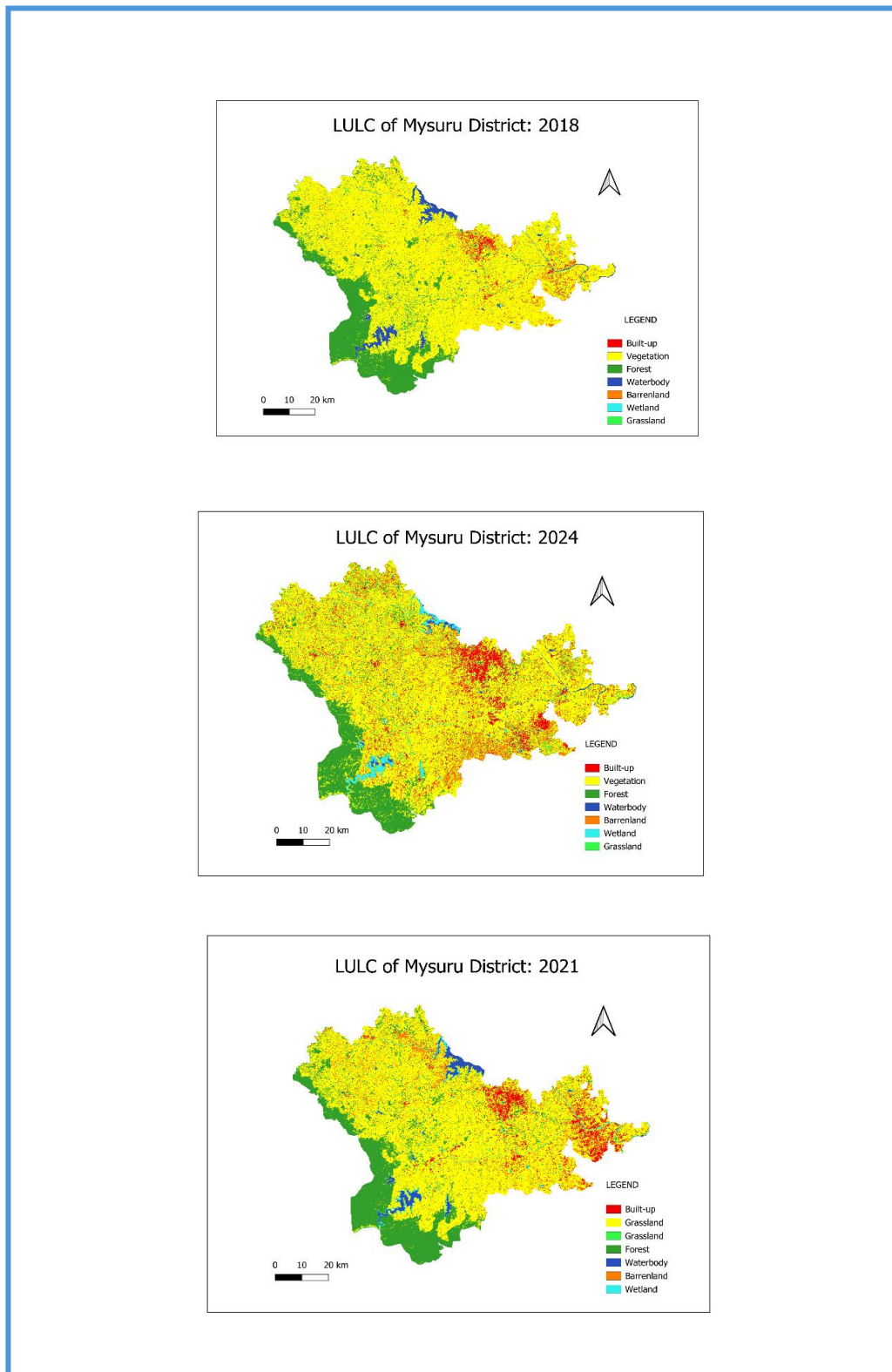


Fig 6: LULC Map layouts of Mysuru District using Random Forest (2018, 2021, 2024)

YEARS	OVERALL ACCURACY	KAPPA COEFFICIENT
2018	92.85%	90.75%
2021	85.71%	82.56%
2024	94.23%	92.72%

CLASSES	2018		2021		2024	
	PRODUCER ACCURACY	USER ACCURACY	PRODUCER ACCURACY	USER ACCURACY	PRODUCER ACCURACY	USER ACCURACY
Built-Up	100%	100%	87%	100%	100%	94%
Vegetation	100%	83%	77%	100%	100%	92%
Forest	100%	100%	100%	91%	100%	100%
Waterbody	100%	75%	100%	25%	83%	100%
Barren Land	50%	100%	100%	75%	75%	100%
Wetland	50%	100%	0%	0%	100%	85%
Grassland	100%	100%	100%	100%	75%	100%

Table 1: Accuracy assessment of Random Forest model

SUPERVISED CLASSIFICATION USING SVM (GEE)

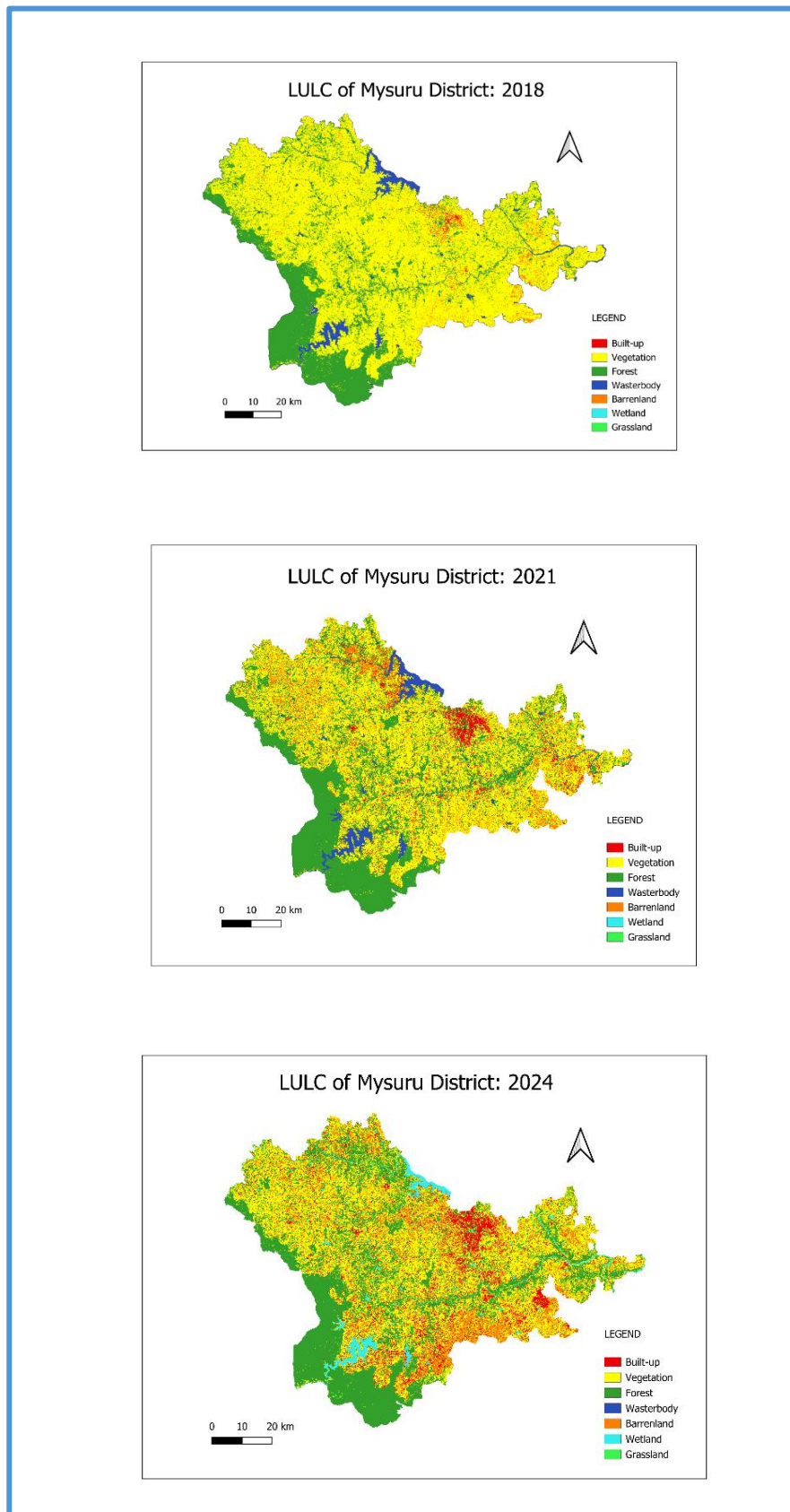


Fig 7: LULC Map layout of Mysuru District using SVM (2018,2021,2024)

YEARS	OVERALL ACCURACY	KAPPA COEFFICIENT
2018	76.19%	68.46%
2021	73.80%	68.11%
2024	82.69%	78.31%

Classes	2018		2021		2024	
	Producer Accuracy	User Accuracy	Producer Accuracy	User Accuracy	Producer Accuracy	User Accuracy
Built-up	60%	100%	75%	85.71%	100%	100%
Agricultural Land	90%	60%	33.33%	75%	100%	92.85%
Forest	100%	93.75%	100%	84.61%	100%	75%
Waterbodies	100%	60%	100%	25%	0%	0%
Wastelands	50%	66.66%	100%	60%	100%	80%
Wetlands	0%	0%	0%	0%	100%	50%
Grass/Grazing land	0%	0%	100%	100%	25%	100%

Table 2: Accuracy assessment of SVM model

SUPERVISED CLASSIFICATION USING ERDAS IMAGINE

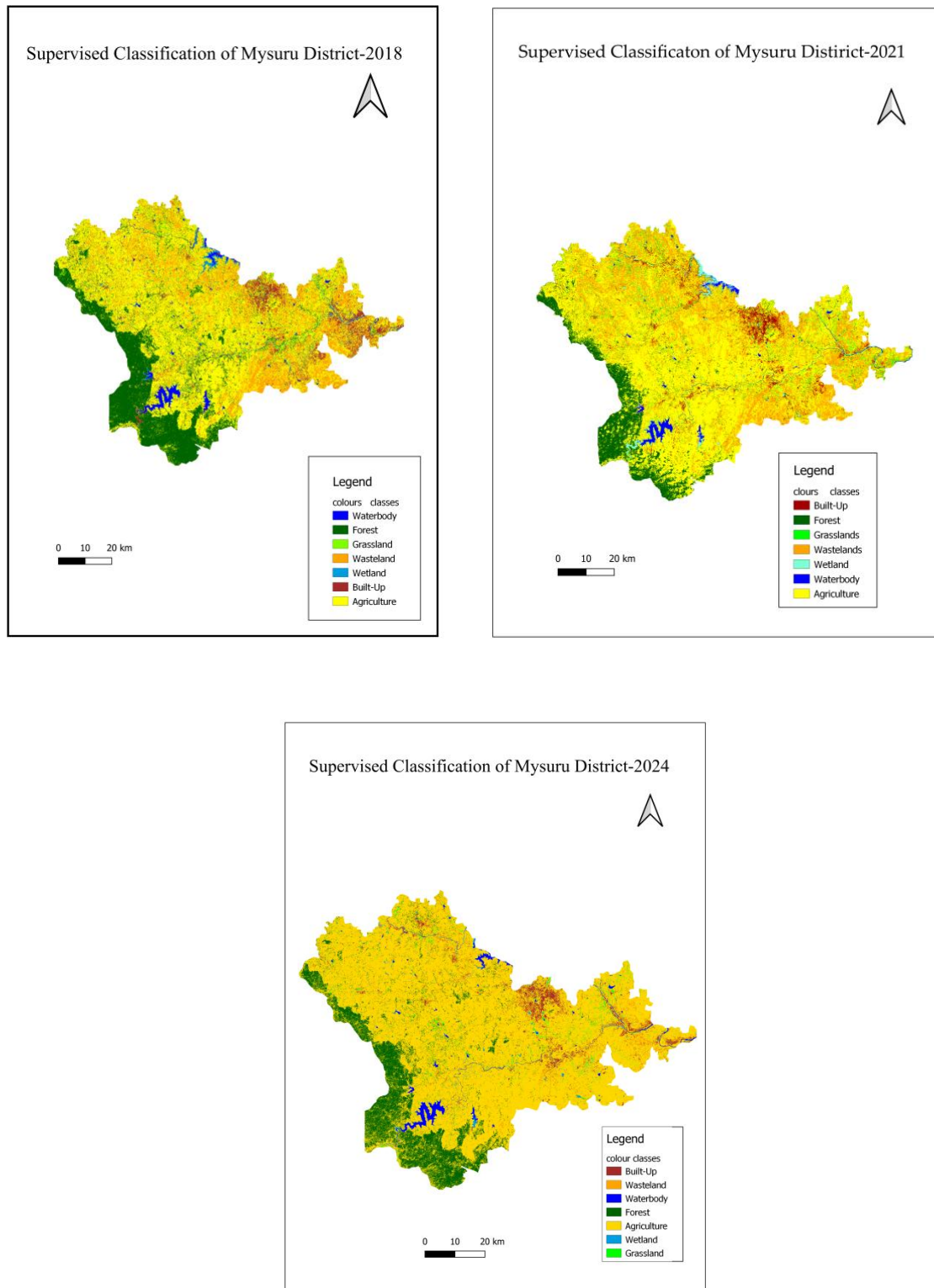


Fig 8: LULC Map layouts of Mysuru District using SVM (2018, 2021, 2024)

YEARS	OVERALL ACCURACY	KAPPA COEFFICIENT
2018	76.67%	67.79%
2021	83.33%	70.00%
2024	91.30%	88.73%

Classes	2018		2021		2024	
	Producer Accuracy	User Accuracy	Producer Accuracy	User Accuracy	Producer Accuracy	User Accuracy
Built-up	100%	100%	0%	0%	100%	88.89%
Agricultural Land	90.91%	62.50%	100%	100%	100%	100%
Forest	85.71%	100%	0%	0%	100%	100%
Grass/Grazing land	0%	0%	80%	66.67%	0%	0%
Wastelands	80%	80%	0%	0%	50%	100%
Wetlands	100%	100%	100%	100%	100%	100%
Waterbodies	0%	0%	85%	94.44%	50%	100%

Table 3: Accuracy assessment using ERDAS IMAGINE

AREA STATISTICS

VALUE	CLASSES	AREA (SQ. KM)		
		2018	2021	2024
0	Built-Up	222.68	398.54	640.7
1	Vegetation	4396.58	4054.75	3901.5
2	Forest	1268.19	1323.66	948.7
3	Waterbody	178.80	125.57	51.2
4	Barren Land	130.55	270.66	324.5
5	Wetland	32.76	91.20	118.8
6	Grassland	82.31	47.49	326.5

Table 4: Area Statistics (sq. km)

2.5 INTERPRETATION

The spatio-temporal Land Use Land Cover (LULC) analysis of Mysuru District from 2018 to 2024 shows significant changes in land cover patterns due to urban expansion, environmental changes, and human activities.

The area statistics indicate a continuous increase in built up land from 222.68 sq. km in 2018 to 640.7 sq. km in 2024, showing rapid urban growth and infrastructure development in the district. At the same time, vegetation cover decreased from 4396.58 sq. km to 3901.5 sq. km, while forest area reduced from 1268.19 sq. km to 948.7 sq. km. This reflects the conversion of natural and agricultural landscapes into urban and other land use categories. Waterbody area also declined significantly from 178.80 sq. km in 2018 to 51.2 sq. km in 2024, indicating possible impacts of reduced rainfall, land conversion, and environmental stress.

Barren land, wetlands, and grasslands showed noticeable fluctuations during the study period. Barren land increased from 130.55 sq. km to 324.5 sq. km, suggesting land degradation and expansion of

unused land surfaces. Wetlands increased gradually from 32.76 sq. km to 118.8 sq. km, while grassland increased sharply in 2024 to 326.5 sq. km, which may indicate seasonal variation, classification differences, or changes in land management practices.

The Random Forest classification in Google Earth Engine produced the highest classification accuracy among all methods. The overall accuracy values were 92.85% in 2018, 85.71% in 2021, and 94.23% in 2024 with strong Kappa coefficient values above 82%, indicating reliable and consistent classification performance.

The SVM classification also produced acceptable results but with comparatively lower accuracy values of 76.19%, 73.80%, and 82.69% for 2018, 2021, and 2024 respectively. Certain classes such as wetlands and waterbodies showed low producer and user accuracy in some years, indicating spectral confusion among land cover classes.

The ERDAS Imagine supervised classification demonstrated improved performance over time, achieving 91.30% overall accuracy and 88.73% Kappa coefficient in 2024. However, some classes showed inconsistent producer and user accuracy values, especially in 2021 and 2024, which suggests limitations in sample selection and class separability.

CHAPTER 3

CONCLUSION AND REFERENCES

3.1 CONCLUSION

The study successfully analysed the spatiotemporal Land Use Land Cover changes in Mysuru District between 2018 and 2024 using Google Earth Engine and ERDAS Imagine. The results clearly indicate rapid urban expansion and a reduction in vegetation, forest cover, and waterbody areas over the study period. These changes reflect increasing human influence, urbanisation, and environmental pressure within the district.

Among the classification techniques, the Random Forest algorithm provided the highest classification accuracy and better reliability compared to SVM and ERDAS Imagine supervised classification. The integration of machine learning techniques with remote sensing and GIS tools proved highly effective for LULC mapping and change detection.

The study highlights the importance of continuous environmental monitoring for sustainable land resource management and urban planning. The generated LULC maps and statistical analysis can support government agencies, planners, and researchers in understanding land transformation patterns and making informed decisions for environmental conservation and regional development.

3.2 REFERENCES

Breiman, L., 'Random Forests', *Machine Learning*, 45.1 (2001), pp. 5–32.

Congalton, R.G. and Green, K., *Assessing the Accuracy of Remotely Sensed Data: Principles and Practices*, 2nd edn (Boca Raton: CRC Press, 2009).

Foody, G.M., 'Supervised Image Classification by MLP and RBF Neural Networks with and without an Exhaustive Search for the Appropriate Architecture', *International Journal of Remote Sensing*, 22.16 (2001), pp. 3091–3104.

Foody, G.M., 'Status of Land Cover Classification Accuracy Assessment', *Remote Sensing of Environment*, 80.1 (2002), pp. 185–201.

Google Earth Engine Team, *Google Earth Engine: Planetary-Scale Geospatial Analysis for Everyone* <https://earthengine.google.com/> [accessed 24 May 2026].

Jensen, J.R., *Introductory Digital Image Processing: A Remote Sensing Perspective*, 4th edn (Boston: Pearson Education, 2015).

Lillesand, T., Kiefer, R.W. and Chipman, J., *Remote Sensing and Image Interpretation*, 7th edn (New York: Wiley, 2015).

Richards, J.A. and Jia, X., *Remote Sensing Digital Image Analysis: An Introduction*, 5th edn (Berlin: Springer, 2006).

Stehman, S.V., 'Selecting and Interpreting Measures of Thematic Classification Accuracy', *Remote Sensing of Environment*, 62.1 (1997), pp. 77–89.

ERDAS Inc., *ERDAS IMAGINE 2014 Field Guide* (Norcross, Georgia: Hexagon Geospatial, 2014).

Maxwell, A.E., Warner, T.A. and Fang, F., 'Implementation of Machine-Learning Classification in Remote Sensing: An Applied Review', *International Journal of Remote Sensing*, 39.9 (2018), pp. 2784–2817.

Pal, M. and Mather, P.M., 'An Assessment of the Effectiveness of Decision Tree Methods for Land Cover Classification', *Remote Sensing of Environment*, 86.4 (2003), pp. 554–565.

Belgiu, M. and Drăguț, L., 'Random Forest in Remote Sensing: A Review of Applications and Future Directions', *ISPRS Journal of Photogrammetry and Remote Sensing*, 114 (2016), pp. 24–31.

Mountrakis, G., Im, J. and Ogole, C., 'Support Vector Machines in Remote Sensing: A Review', *ISPRS Journal of Photogrammetry and Remote Sensing*, 66.3 (2011), pp. 247–259

